<Company Name>

# Arithmetic Expression Evaluator
# Software Development Plan
**Version v0.1**

# Revision History

| Date | Version | Description | Author(s) |
|---|---|---|---|
| <2/22/2026> | 0.1 | Team member rolls assigned, project schedule defined, meeting times stated, managerial systems outlined. | Dylan Richardson, Gabriel Gonzalez, Kevin Reiley, Jack Rummell, Eli Chapman, Lawrence Burton III |
| <3/15/2026> | 0.2 | Role changes, schedule further defined, other small changes | Jack Rummell |
| | | | |
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# Software Development Plan

## 1. Introduction

Our team will be designing, developing, and testing an arithmetic expression evaluator. This document will serve as the central source of information for all parties involved in the project. The project manager will use this document to organize information and team members will use it to answer questions and find clarity about aspects of the project.

### 1.1 Purpose

The purpose of the *Software Development Plan* is to organize all information necessary to control the project. It is the top-level plan generated and used by managers to direct the development effort. It outlines the schedule, lists useful documents, describes the development approach, and contains information about the team such as who we are, what each members role is, and how to contact them.

The following people use the *Software Development Plan*:

- The **project manager** uses it to plan the project schedule and resource needs, and to track progress against the schedule.
- **Project team members** use it to understand what they need to do, when they need to do it, and what other activities they are dependent upon.

### 1.2 Scope

This Software Development Plan describes the overall plan to be used by the arithmetic expression evaluator project, including deployment of the product. The details of the individual iterations will be described in the Iteration and Phase Documentation.

The plans as outlined in this document are based upon the product requirements as defined in the Vision Document

### 1.3 Definitions, Acronyms, and Abbreviations

SRS – Software Requirements Specification

SAS – Software Architecture Specification

TCS – Test Case Specification

Key Member – A team member with more responsibility. Key members are dependent on the current iteration and outlined in the iteration and phase documentation.

### 1.4 References

- Iteration and Phase Documentation
- Software Requirements Specification (SRS)
- Software Architecture Specification (SAS)
- Test Case Specification (TCS)
- User Manual

If you are a team member trying to get access to any documents, first check the github repo, then reach out to Jack Rummell via slack.

### 1.5 Overview

This *Software Development Plan* contains the following information:

Project Overview — provides a description of the project's purpose, scope, and objectives. It also defines the deliverables that the project is expected to deliver.

Project Organization — describes the organizational structure of the project team.

Management Process — explains the estimated cost and schedule, defines the major phases and milestones for the project, and describes how the project will be monitored.

Applicable Plans and Guidelines — provide an overview of the software development process, including methods, tools and techniques to be followed.

## 2. Project Overview

### 2.1 Project Purpose, Scope, and Objectives

The purpose of this project is to create a program that parses and evaluates basic arithmetic expressions. Such expressions can include standard operations such as addition, subtraction, multiplication, division, and exponentiation. Further specifications are outlined in the SRS.

The main objectives of this project consist of deliverables (Section 2.3). This project is expected to deliver these deliverables, including the templates used in developing the software, and the source code itself. These deliverables will be of high quality and ensure the final product validates all requirements.

### 2.2 Assumptions and Constraints

Assumptions:

- The final product of this project will be instantiated and used by a compiler

Constraints:

- Time

  Team members are full-time students and have other activities, such as jobs, they must attend too.

- Experience

  Team members have varying degrees of experience with different tooling/languages which could cause learning delays.

- Team size

  Our team is limited to six members. This means deliverables and task allocation must be reasonable for a six-member team.

- C++

  Our project must be written in C++

### 2.3 Project Deliverables

Deliverables are delivered towards the end of the iteration, as specified in section *4.1.3 Project Schedule*.

The deliverables include the following:

- Software Development Plan (Due 2/22/2026)
- Requirements Specification (Due 3/15/2026)

- Software Architecture (Due 4/5/2026)
- Implementation, Test Cases, and User Manual (Due 5/7/2026)

### 2.4 Evolution of the Software Development Plan

The *Software Development Plan* will be revised prior to the start of each Iteration during the planning phase. These revisions will be noted and given brief summaries within each new revised version of this document. Depending on the nature of development, and the expected lack of changes in requirements, it is likely there will be few, if any, revisions to this plan in the future.

## 3. Project Organization

### 3.1 Organizational Structure

The organizational structure of this project is broken into several roles, each with various responsibilities and expected contributions to the development process.

| Team Roles | |
| --- | --- |
| **Team Member** | **Unified Process for EDUcation Role** |
| Gabriel Gonzalez | Analyst |
| Eli Chapman | Designer |
| Dylan Richardson, Kevin Reilly | Implementer |
| Kevin Reilly | Integrator |
| Dylan Richardson | Tester |
| Jack Rummell | Change Control Manager |
| Lawrence Burton III | Configuration Manager |
| Jack Rummell | Project Manager |
| Lawrence Burton III | Reviewer |
| Gabriel Gonzalez | Any Roll |
| Full Team | Stakeholder |

| Availability | | | | | | | |
| --- | --- | --- | --- | --- | --- | --- | --- |
| **Team Member** | **Sunday** | **Monday** | **Tuesday** | **Wednesday** | **Thursday** | **Friday** | **Saturday** |
| Jack Rummell | N/A | 3:20pm – 6:00pm | 3:35pm - 6:00pm | N/A | 3:35pm – 6:00pm | 11:00am – 6:00pm | N/A |
| Dylan Richardson | N/A | 3:30pm – 6:00pm | 10:30am – 2:00pm | 10:00am – 5:00pm | N/A | 8:00am – 5:00pm | N/A |
| Eli Chapman | N/A | 3:30pm – 8:00pm | 3:30pm – 8:00pm | 3:30pm – 8:00pm | 3:30pm – 8:00pm | 3:30pm – 8:00pm | N/A |
| Gabriel Gonzalez | N/A | 1:30pm-10pm | 3:40pm-10pm | 3:30pm-10pm | N/A | 12pm-8pm | N/A |
| Kevin Reilly | N/A | N/A | 4pm - 10pm | 11:30am – 4pm, 7pm - 10pm | 4pm - 10pm | N/A | N/A |

| | | | | | | | |
| --- | --- | --- | --- | --- | --- | --- | --- |
| Lawrence Burton III | | 3:15pm | | | | | |

Communication between team members will be handled primarily by slack and email. Other communication channels should be used rarely if at all.

| Team Info | | | |
| --- | --- | --- | --- |
| **Team Member** | **Phone** | **Email** | **Skills** |
| Jack Rummell | (785) 727-6090 | J071r165@ku.edu | WebDev, C++, Java |
| Dylan Richardson | (913) 302-7366 | D347r606@ku.edu | C++, Java, C#, WebDev, CMake |
| Eli Chapman | (785) 409-7620 | eychapman@ku.edu | C++, Dart, WebDev, React TS, Python |
| Gabriel Gonzalez | (785) 304-1437 | G203g367@ku.edu | Python |
| Kevin Reilly | (913) 424-3781 | k156r438@ku.edu | Java, Javascript, HTML &CSS, SQL, UNIX |
| Lawrence Burton III | (913) 475-0848 | Lawrencewburtoniii@ku.edu or lawrence.burton@catalyst-studios.cc | Python, C++, C#, Flutter, Dart, Ruby, |

# 4. Management Process

## 4.1 Project Plan

### *4.1.1 Iteration Objective*

The project will be broken down into four iterations. Each iteration will consist of three phases. For a comprehensive overview of phases and iterations, reference the iteration and phase documentation.

Iterations:

Inception – The goal of this iteration is to establish a requirements overview

Elaboration – The goal of this iteration is to plan the software architecture and refine the requirements overview

Construction – The goal of this iteration will be to develop the code necessary for the project

Transition – The goal of this iteration is to validate that the implementation meets all requirements and build the user manual

Phases:

Planning phase – The goal of this phase is to determine and allocate tasks to team members.

Building phase – The goal of this phase is to work down tasks and build needed artifacts.

Quality control phase – The goal of this phase is to ensure the project is ready to progress to the next iteration.

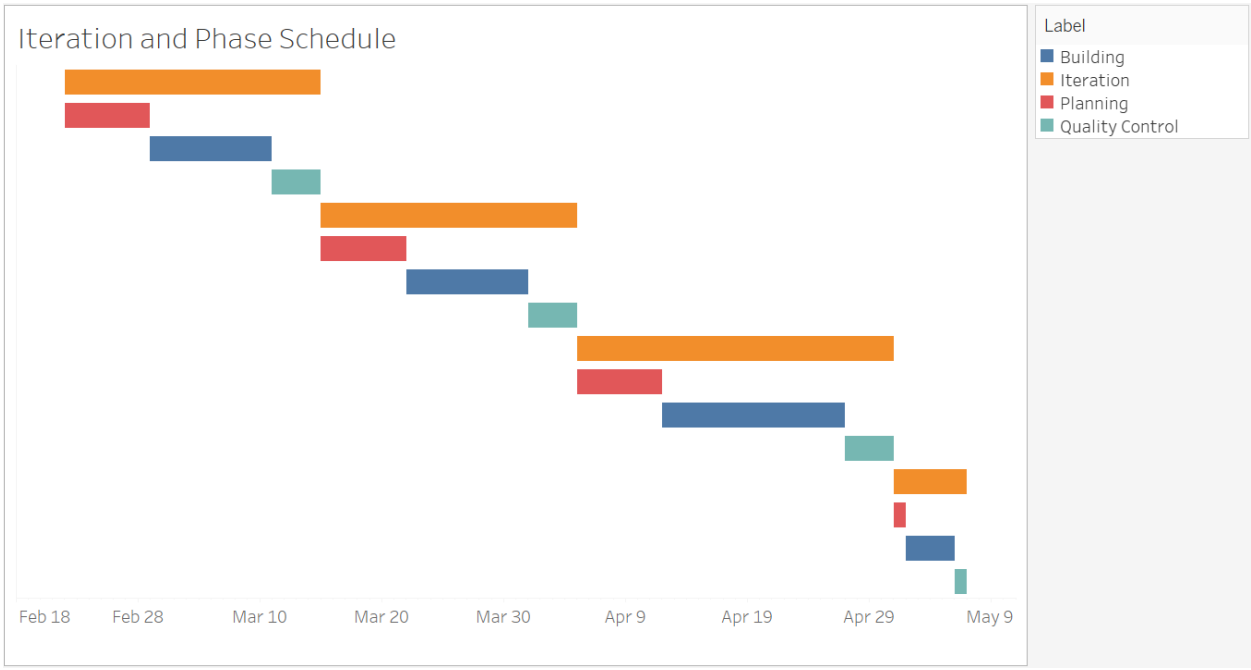
### *4.1.2 Releases*

There will be a single release. It will contain an arithmetic expression evaluator with all requirements tested

and validated. The release will be accompanied by a user manual.

### 4.1.3 *Project Schedule*

| Deliverables | |
| --- | --- |
| **Milestone** | **Target Completion Date** |
| Software Requirements Specification | 3/15/2026 |
| Software Architecture Specification | 4/05/2026 |
| Implementation Artifacts | 5/01/2026 |
| Test Case Specification | 5/07/2026 |
| User Manual | 5/07/2026 |




### 4.1.4 *Meeting Details*

Our team will meet weekly in person or via zoom. Since our schedules don't share availability any day of the week, our weekly meetings will be dependent on the current iteration. Meetings times are based on the availability of the key members of the current iteration.

Meeting dates are tentative.

Inception:
Tuesdays @ 4:00pm
Elaboration:
Tuesdays @ 4:00pm
Construction:
Tuesdays @ 4:00pm

Transition:

Tuesdays @ 4:00pm

The project manager will maintain meeting logs and take notes to be shared with the full team post-meeting.

#### *4.1.5 Task Management and Accountability*

Tasks will be allocated via slack. Each team member will have a dedicated channel the project manager will use to list tasks and completion dates.

The project manager will maintain a spreadsheet of all tasks, along with information about each task such as the assigned team member, due date, completion date, etc. Tasks not completed or not completed by the completion date will be flagged and noted. The spreadsheet will be turned in with the final submission of the term project for Professor Saiedian to do with as he pleases.

### 4.2 Project Monitoring and Control

#### *4.2.1 Quality Control*

Defects will be recorded and tracked as Change Requests, and defect metrics will be gathered. Additionally, each iteration will have a quality control phase which ensures quality of any artifacts created in the iteration.

All deliverables are required to go through the appropriate review process, as described in the Development Case. The review is required to ensure that each deliverable is of acceptable quality, using guidelines and checklists.

Any defects found during review which are not corrected prior to releasing for integration must be captured as Change Requests so that they are not forgotten.

#### *4.2.2 Risk Management*

Risks will be identified in Inception Phase using the steps identified in the RUP for Small Projects activity “Identify and Assess Risks”. Project risk is evaluated at least once per iteration and documented in this table.

#### *4.2.3 Configuration Management*

Appropriate tools will be selected which provide a database of Change Requests and a controlled versioned repository of project artifacts.

All source code, test scripts, and data files are included in baselines. Documentation related to the source code is also included in the baseline, such as design documentation. All customer deliverable artifacts are included in the final baseline of the iteration, including executables.

The Change Requests are reviewed and approved by one member of the project, the Change Control Manager role.

## 5. Annexes

The project will follow the UPEDU process.

Programming guidelines:

All source files and header files will start with the following comment template…

/*

Authors:

Description:

*/